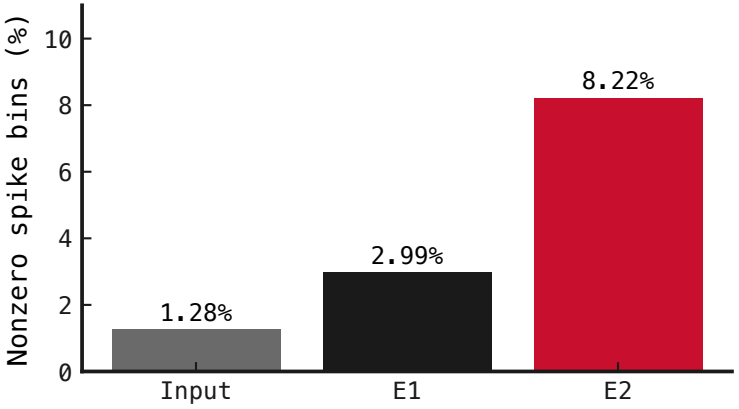
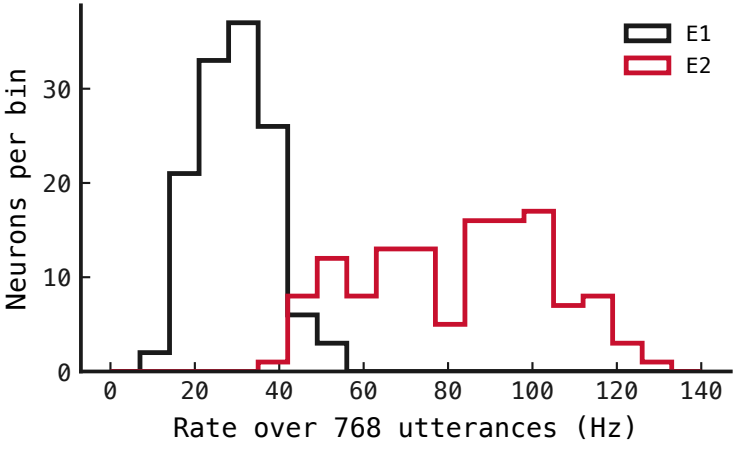
SHD / sparse spikes, dense sampled gradients

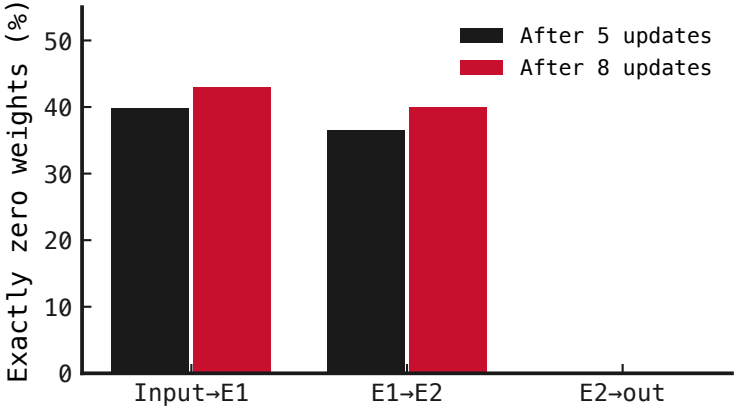
A Spike density



B Per-neuron rates



C Numerical weight zeros



D Block and gradient structure

Projection	Zero 16×16 tiles
Input→E1	0/352
E1→E2	0/64
E2→out	0/16

100% of sampled gradient entries nonzero in all three learned matrices (three batches; before clipping).

Diagnostic batches 6–8 of a fresh run | early training, not the final checkpoint